

🎵 DEEP LEARNING

📊 GTZAN · RESNET-18

Deep Learning for Music Genre Classification

Converting raw audio into log-scaled Mel-spectrograms and applying transfer learning with a pre-trained ResNet-18 to classify 10 musical genres from the GTZAN dataset - achieving **94.06% validation accuracy**, a significant improvement over the prior CNN benchmark of ~80%.

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Background & Motivation

Music genre classification assigns labels like "Jazz" or "Rock" to audio tracks automatically, with applications in recommendation, organisation, and analysis. Human listeners distinguish genres by timbre, rhythm, and instrumentation, but automatic classification remains challenging due to genre ambiguity and variability. Previous research using CNNs on GTZAN achieved around 80% test accuracy, indicating substantial room for improvement.

Core Problem

Feeding raw 30-second clips or full-length spectrograms directly into a basic CNN on only 1,000 samples results in severe overfitting, initial models memorised training data but failed to generalise to unseen data.

Our Approach

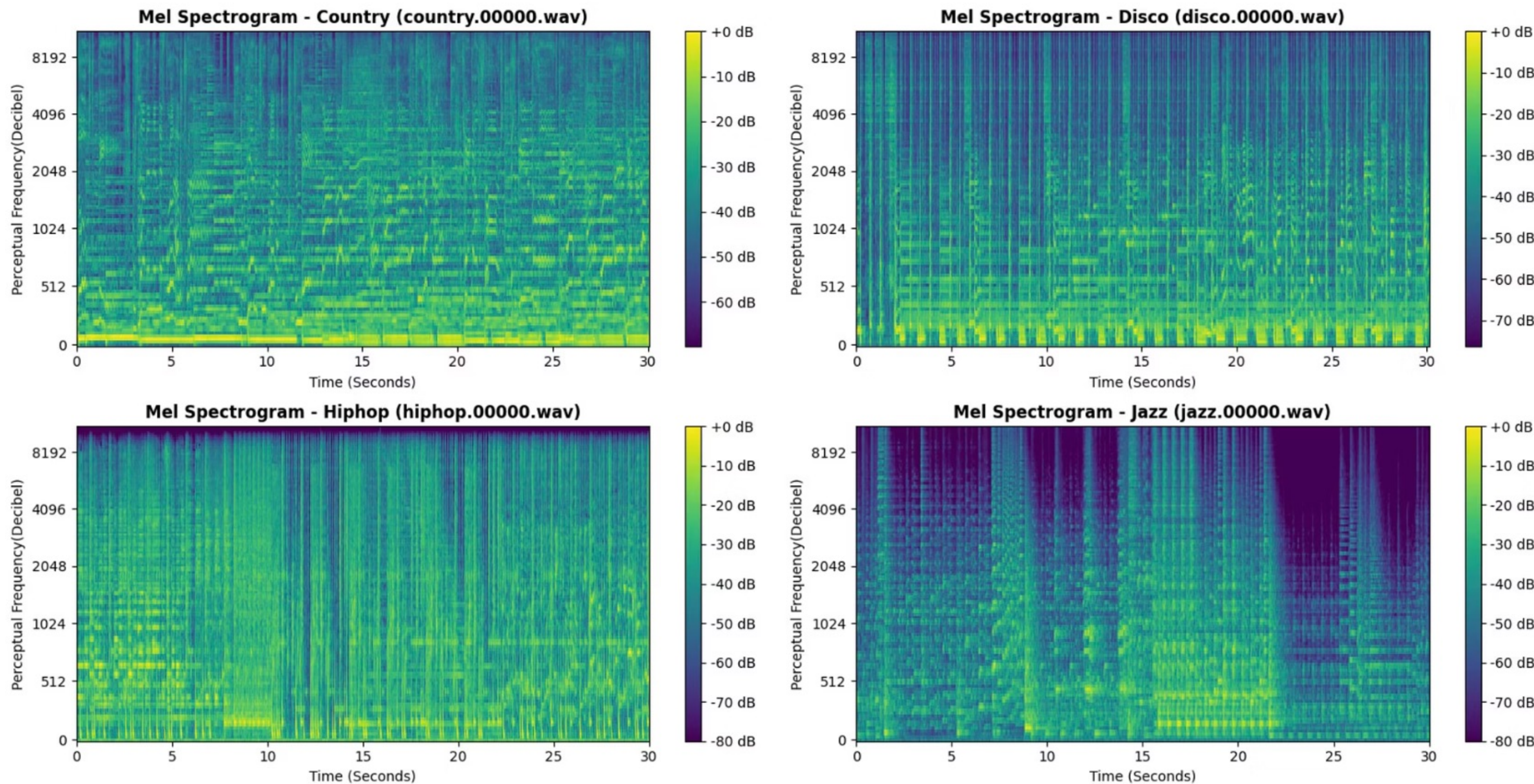
We systematically resolve overfitting through data augmentation (3-second slicing) and transfer learning (ResNet-18), treating audio classification as a computer vision problem on Mel-spectrogram images.

Dataset & Preprocessing

The GTZAN dataset often called the "MNIST of sounds" contains 1,000 audio excerpts (30 seconds each) evenly split across 10 genres: Blues, Classical, Country, Disco, Hip Hop, Jazz, Metal, Pop, Reggae, and Rock. Each genre has 100 tracks, providing a controlled benchmark for classifier development.

01	02	03
Mel-Spectrogram Extraction Each audio file is loaded via Librosa, resampled to a standard rate, and converted to a mel-spectrogram with 128 mel bands, a 2D image representing power distribution over time and mel-frequency bins. Spectrograms are converted to decibel (log) scale to mimic human perception of loudness.	Data Augmentation via Slicing Each 30-second spectrogram (~128×1290) is sliced into ten 3-second frames (~128×129), increasing the dataset tenfold to ~10,000 samples. Each slice is standardised with per-band normalisation. This approach, validated by Nanni et al. (2020), acts as a powerful regulariser against overfitting.	Train / Validation / Test Split 15% reserved for test, then the remaining 85% split into ~70% training and 15% validation. This ensures robust evaluation while maximising training data availability.

Mel-Spectrogram Extraction



Model Architecture & Training

The project progressed through three sequential models, each addressing the limitations of its predecessor:



Basic CNN

A 3-layer 2D CNN mapping full 1290-width spectrograms. Training accuracy reached 97.28% but test accuracy was only 58.00% - severe overfitting due to insufficient data.



Optimised CNN

Used the sliced dataset (128×129 input) with Batch Normalisation, increased Dropout (0.5), Weight Decay (1e-4), and Early Stopping (patience = 10). Test accuracy improved to 83.92%.



Transfer Learning (ResNet-18)

A pre-trained ResNet-18 (ImageNet) with the final FC layer modified to output 10 classes. Single-channel spectrograms duplicated across 3 channels to mimic RGB. Fine-tuned with a low learning rate - achieving **94.06% validation accuracy**.

All models are implemented in PyTorch. The optimizer is Adam with a learning rate of 0.001. L2 weight decay (1e-4) penalises large weights. Early stopping halts training if validation loss does not improve for several consecutive epochs, saving the best model weights. Training runs for up to 50 epochs with a batch size of 32 on a GPU via Google Colab.

Results & Discussion

94.06%

Validation Accuracy
Final ResNet-18 model

~13%

Improvement
Over prior CNN benchmark
(~80%)

10K

Training Samples
From 1,000 clips × 10 slices

10

Genres Classified
Blues through Rock

Training loss quickly falls toward zero with training accuracy reaching ~99.6%. Validation loss generally decreases but exhibits slight fluctuations after epoch 15, indicating the onset of overfitting. Early stopping was triggered at peak validation accuracy (~93.4%), with the model converging stably within 15-20 epochs.

On the test set, most genres are classified with very high accuracy (often >95% per genre). Classical and Jazz tracks are almost perfectly identified (99-100% correct). Notable confusions include:

Rock ↔ Country/Blues

8 Country tracks mistaken for Rock;
6 Rock tracks for Blues or Metal -
shared guitar timbre and rhythm
patterns.

Pop ↔ Disco/Reggae

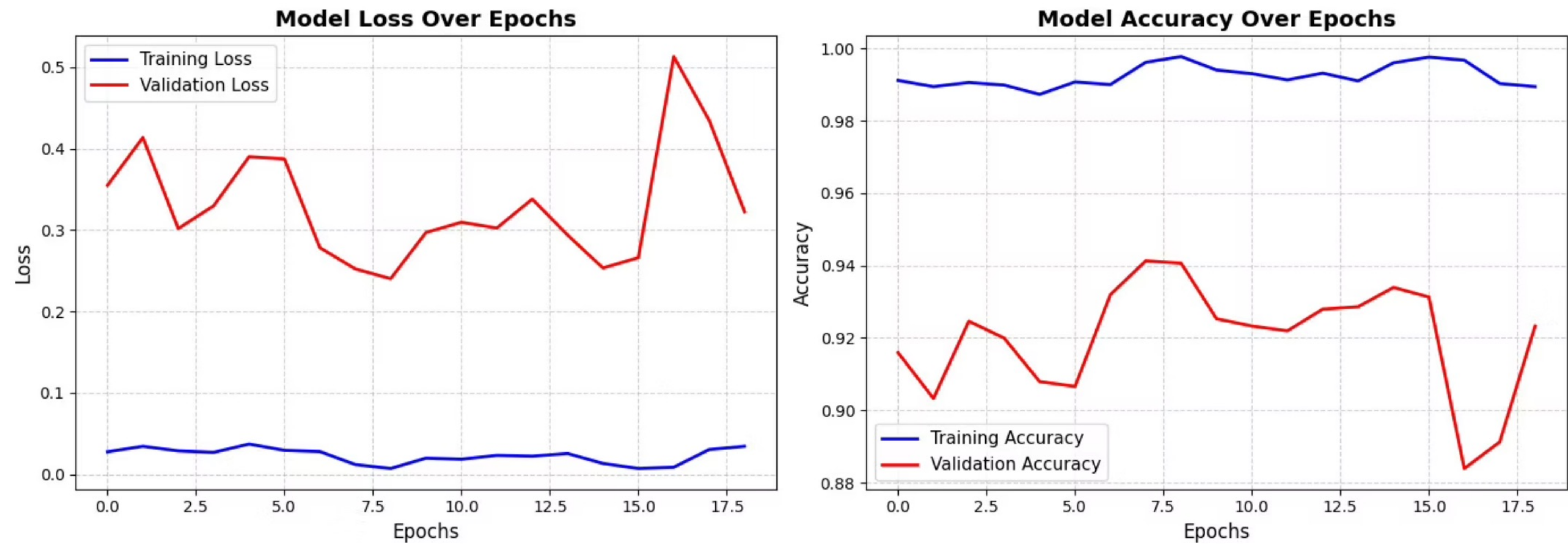
2-3% of Pop samples confused as Hip
Hop or Disco; some Reggae tracks as
Pop or Country - shared rhythmic
features.

Metal ↔ Rock

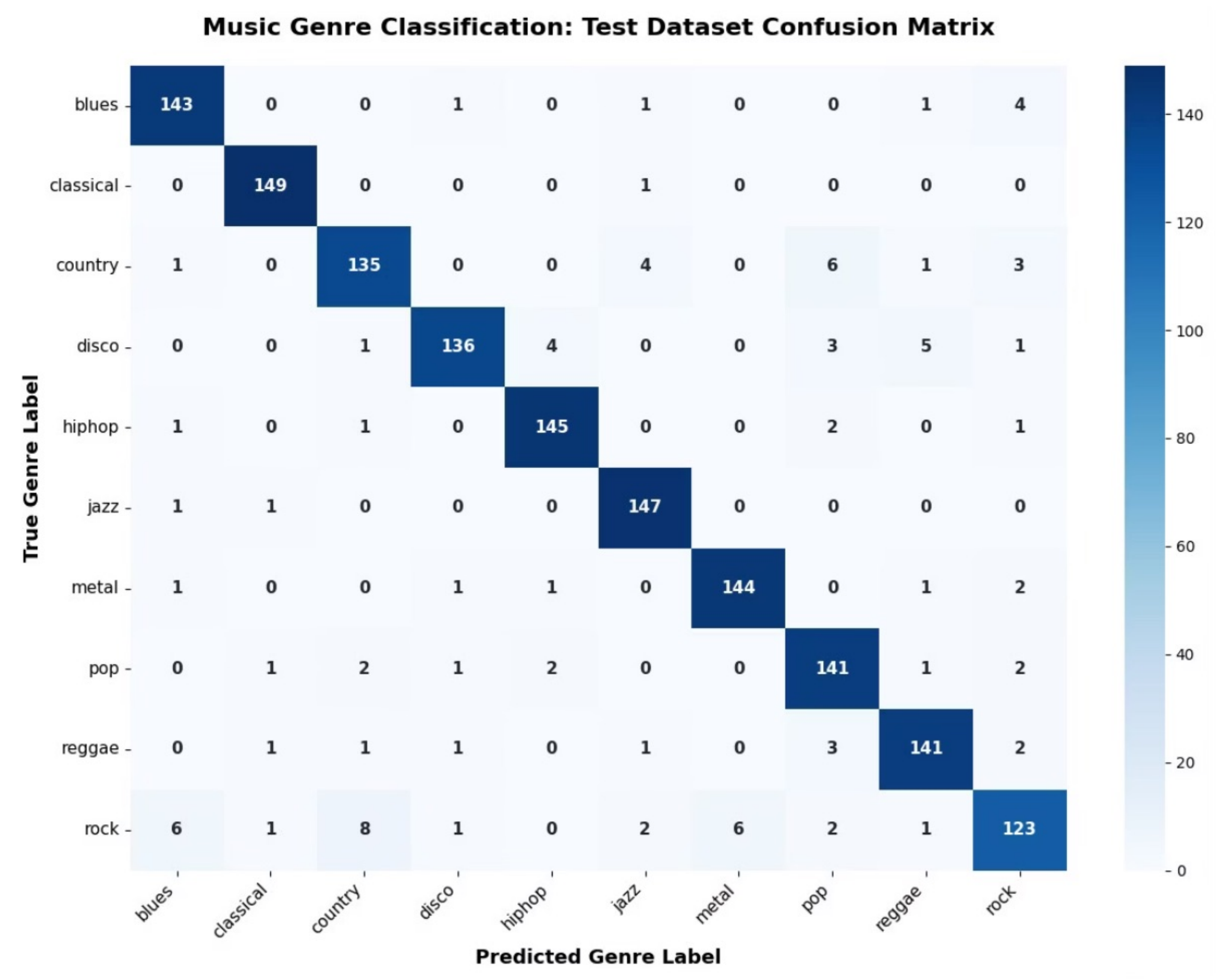
2 out of 150 Metal tracks
misclassified as Rock - overlapping
energy profiles and distortion-
dominated sound.

These misclassifications align with intuitive audio overlap: Rock/Country/Blues all feature prominent guitars; Pop and Disco share upbeat rhythms; Reggae and Pop share syncopated beats. The confusion matrix confirms strong per-genre performance, with diagonal entries mostly near 140-149 out of 150 and all off-diagonals ≤ 8 .

Loss & Accuracy VS Epochs

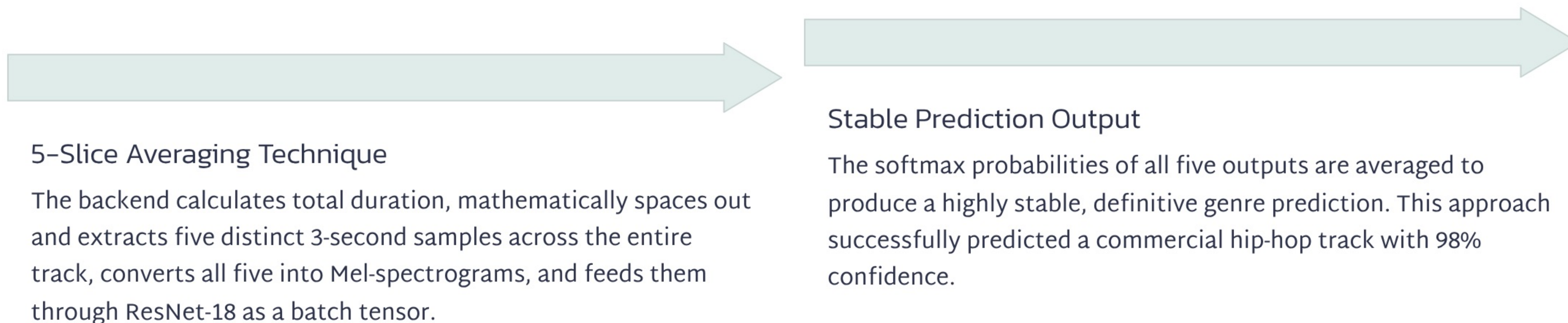


Confusion Matrix



Real-World Application

A functional web application was developed using Gradio to allow users to upload custom audio files for prediction. Testing real-world, multi-minute audio highlighted a significant statistical limitation: the model was trained exclusively on 3-second chunks from a small pool of 1,000 original clips. Relying on a single 3-second slice is risky - predicting based on a quiet intro or anomalous bridge.



Real-Time Unseen Music Classification

Music Genre Classifier Pro

This model samples 5 segments from your song to ensure a highly accurate genre prediction.

Upload Music File

Uploading 1 file...

50_Cent_-_In_Da_Club__Official_Music_Video_(256k).mp3

Clear

Submit

Flag

Top 3 Predicted Genres

hiphop

hiphop 98%

pop

pop 2%

reggae

reggae 0%

Conclusion & Future Work

This project successfully developed a high-accuracy music genre classifier by treating audio processing as a computer vision problem. By converting raw audio waves into log-scaled Mel-spectrograms, augmenting the limited GTZAN dataset via 3-second slicing, and applying transfer learning through a pre-trained ResNet-18, the final system achieved 94.06% validation accuracy. The 5-slice averaging algorithm further enabled reliable processing of variable-length, real-world audio files.

Key Challenges

- **Dataset Scarcity:** 1,000-clip GTZAN is extremely small for deep learning, necessitating aggressive augmentation to prevent memorisation.
- **Subjectivity in Music:** Songs with cross-genre influences naturally confuse categorical neural networks.
- **Input Dimension Constraints:** Adapting a 3-second-snapshot model to full-length songs required algorithmic workarounds at the application level.

Future Directions

- Train on larger, more diverse datasets (e.g., Free Music Archive, Million Song Dataset).
- Explore sequence-based architectures: RNNs or Audio Transformers to learn temporal, long-term dependencies of rhythm and structure.
- Develop real-time demo or mobile app integration for on-device classification.

References

- [1] Tzanetakis, G. & Cook, P. (2002). Musical genre classification of audio signals. IEEE Transactions on Speech and Audio Processing.
- [2] Choi, K. et al. (2016). Convolutional recurrent neural networks for music classification. arXiv preprint.
- [3] Nanni, L. et al. (2020). Data augmentation approaches for improving animal audio classification. Ecological Informatics.
- [4] Gulum, M. (2022). Music Genre Classification. GitHub repository.

 Questions welcome! Thank you for your attention.